MATHEMATICS PART 2 (GEOMETRY) | CLASS 10 | MAHARASHTRA SSC BOARD

# Chapter 5: Coordinate Geometry

**MODEL ANSWER SHEET / ANSWER KEY**

| 43 | 18 | 4 |
|---|---|---|
| QUESTIONS | IN TEXTBOOK | FIGURES |

**HOW TO USE**
*Cover the answer box ✓ and answer first. A* **TEXTBOOK** *strip means the same question is in your Balbharati Math-2 textbook, at the set and page shown. Tick ☐ each section once revised.*

| Q.1 (A) | Q.1 (B) | Q.2 (A) | Q.2 (B) | Q.3 (A) | Q.3 (B) | Q.4 | Q.5 |
|---|---|---|---|---|---|---|---|
| Multiple Choice | One-step Solve | Activities | Short Solve | Activities | Solve / Prove | Long Solve / Prove | Creative / HOTS |
| 10 Q | 10 Q | 4 Q | 5 Q | 2 Q | 7 Q | 3 Q | 2 Q |

## Q.1 (A) Multiple Choice — 1 Mark each

10 QUESTIONS · 6 ☐ Revised

**Q1** Point P is midpoint of segment AB where $A(-4, 2)$ and $B(6, 2)$, then the coordinates of P are _____.

**TEXTBOOK** Solved Example · Ex. (2) · Page 111

(a) $(-1, 2)$ (b) $(1, 2)$ (c) $(1, -2)$ (d) $(-1, -2)$

**✓ ANSWER / SOLUTION**

Point P is midpoint of segment AB where $A(-4, 2)$ and $B(6, 2)$, then the coordinates of P are $(1, 2)$.

**Explanation:**

$A(x_1, y_1) = A(-4, 2)$, $B(x_2, y_2) = B(6, 2)$

Here, $x_1 = -4, y_1 = 2, x_2 = 6, y_2 = 2$

$\therefore$ Coordinates of the midpoint of seg AB

$= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$= \left(\frac{-4+6}{2}, \frac{2+2}{2}\right)$

$= \left(\frac{2}{2}, \frac{4}{2}\right)$

$= (1, 2)$

**Q2** The distance between point $P(2, 2)$ and $Q(5, x)$ is $5$ cm, then the value of $x =$?

(a) 2 (b) 6 (c) 3 (d) 1

**ANSWER / SOLUTION**

The distance between point $P(2, 2)$ and $Q(5, x)$ is $5$ cm, then the value of $x = 6$.

**Explanation:**

Let $P(x_1, y_1) = P(2, 2)$ and $Q(x_2, y_2) = Q(5, x)$

Here, $x_1 = 2, y_1 = 2, x_2 = 5, y_2 = x$

By distance formula,

$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$\therefore 5 = \sqrt{(5 - 2)^2 + (x - 2)^2}$

$\therefore 5 = \sqrt{9 + x^2 - 4x + 4}$

$\therefore 5^2 = x^2 - 4x + 13$ ---[Squaring both sides]

$\therefore 25 = x^2 - 4x + 13$

$\therefore x^2 - 4x + 13 - 25 = 0$

$\therefore x^2 - 4x - 12 = 0$

$\therefore (x - 6)(x + 2) = 0$

$\therefore x - 6 = 0$ or $x + 2 = 0$

$\therefore x = 6$ or $x = -2$

**Q3** The distance between points $P(-1, 1)$ and $Q(5, -7)$ is _____.

TEXTBOOK **Solved Example · Ex. (1) · Page 104**

(a) 11 cm (b) 10 cm

(c) 5 cm (d) 7 cm

**ANSWER / SOLUTION**

The distance between points $P(-1, 1)$ and $Q(5, -7)$ is $10$ cm.

**Explanation:**

Let $P(x_1, y_1) = P(-1, 1)$ and $Q(x_2, y_2) = Q(5, -7)$

Here, $x_1 = -1, y_1 = 1, x_2 = 5, y_2 = -7$

By distance formula,

$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{[5 - (-1)]^2 + (-7 - 1)^2}$

$= \sqrt{36 + 64}$

$= \sqrt{100}$

$= 10$ cm

**Q4** If the length of the segment joining point $L(x, 7)$ and point $M(1, 15)$ is $10$ cm, then the value of $x$ is _____.

TEXTBOOK **Practice Set 5.1 · Q.7 · Page 108**

(a) $7$ (b) $7$ **or** $-5$ (c) $-1$ (d) $1$

**ANSWER / SOLUTION**

If the length of the segment joining point $L(x, 7)$ and point $M(1, 15)$ is $10$ cm, then the value of $x$ is $7$ **or** $-5$.

**Explanation:**

Here, $x_1 = x, y_1 = 7, x_2 = 1, y_2 = 15$

By distance formula,

$d(L, M) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$\therefore d(L, M) = \sqrt{(1 - x)^2 + (15 - 7)^2}$

$\therefore 10 = \sqrt{(1 - x)^2 + 8^2}$

$\therefore 100 = (1 - x)^2 + 64$ ---[Squaring both sides]

$\therefore (1 - x)^2 = 100 - 64$

$\therefore (1 - x)^2 = 36$

$\therefore 1 - x = \pm\sqrt{36}$ ---[Taking square root of both sides]

$\therefore 1 - x = \pm 6$

$\therefore 1 - x = 6$ or $1 - x = -6$

$\therefore x = -5$ or $x = 7$

$\therefore$ The value of $x$ is $-5$ or $7$.

**Q5** Find distance between point $A(-3, 4)$ and origin O.

TEXTBOOK **Problem Set 5 · Q.1 (3) · Page 122**

(a) $7$ cm (b) $10$ cm

(c) $5$ cm (d) $-5$ cm

**ANSWER / SOLUTION**

$5$ cm

**Explanation:**

Let $A(x_1, y_1) = A(-3, 4)$ and $O(x_2, y_2) = O(0, 0)$

Here, $x_1 = -3, y_1 = 4, x_2 = 0, y_2 = 0$

By distance formula,

$d(A, O) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{[0 - (-3)]^2 + (0 - 4)^2}$

$= \sqrt{9 + 16}$

$= \sqrt{25}$

$= 5$ cm

**Q6** **If point $P(1,1)$ divide segment joining point A and point $B(-1,-1)$ in the ratio $5:2$, then the coordinates of A are _____.**

(a) $(3,3)$ (b) $(6,6)$ (c) $(2,2)$ (d) $(1,1)$

**ANSWER / SOLUTION**

If point $P(1,1)$ divide segment joining point A and point $B(-1,-1)$ in the ratio $5:2$, then the coordinates of A are $(6,6)$.

**Explanation:**

Let $A(x_1,y_1)$ and $B(x_2,y_2)=B(-1,-1)$

$P(x,y)=P(1,1)$ divides AB in ratio $5:2$

$\therefore x=1, y=1, x_2=-1, y_2=-1, m=5, n=2$

$\therefore$ By section formula,

$x=\frac{mx_2+nx_1}{m+n}$

$\therefore 1=\frac{5(-1)+2x_1}{5+2}$

$\therefore 7=-5+2x_1$

$\therefore 2x_1=7+5$

$\therefore 2x_1=12$

$\therefore x_1=\frac{12}{2}=6$

Similarly, $y_1=6$.

$\therefore$ Coordinates of A are $(6,6)$.

**Q7** **If segment AB is parallel Y-axis and coordinates of A are $(1,3)$, then the coordinates of B are _____.**

TEXTBOOK **Problem Set 5 · Q.1 (1) · Page 122**

(a) $(3,1)$ (b) $(5,3)$ (c) $(3,0)$ (d) $(1,-3)$

**ANSWER / SOLUTION**

If segment AB is parallel Y-axis and coordinates of A are $(1,3)$, then the coordinates of B are $(1,-3)$.

**Explanation:**

Since, seg AB $\parallel$ Y-axis,

$\therefore$ x-coordinate of all points on seg AB will be the same.

x-coordinate of A$(1,3)=1$

x-coordinate of B$(1,-3)=1$

$\therefore (1,-3)$ is the correct option.

**Q8** **If point P is midpoint of segment joining point $A(-4,2)$ and point $B(6,2)$, then the coordinates of P are _____.**

TEXTBOOK **Solved Example · Ex. (2) · Page 111**

(a) $(-1,2)$ (b) $(1,2)$ (c) $(1,-2)$ (d) $(-1,-2)$

**ANSWER / SOLUTION**

If point P is midpoint of segment joining point $A(-4,2)$ and point $B(6,2)$, then the coordinates of P are $(1,2)$.

**Explanation:**

$A(x_1,y_1)=A(-4,2)$, $B(x_2,y_2)=B(6,2)$

Here, $x_1=-4, y_1=2, x_2=6, y_2=2$

$\therefore$ Coordinates of the midpoint of seg AB

$=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$

$=\left(\frac{-4+6}{2},\frac{2+2}{2}\right)$

$=\left(\frac{2}{2},\frac{4}{2}\right)=(1,2)$

**Q9** If point P divides segment AB in the ratio $1:3$ where $A(-5,3)$ and $B(3,-5)$, then the coordinates of P are _____.

(a) $(-2,-2)$ (b) $(-1,-1)$ (c) $(-3,1)$ (d) $(1,-3)$

**ANSWER / SOLUTION**

If point P divides segment AB in the ratio $1:3$ where $A(-5,3)$ and $B(3,-5)$, then the coordinates of P are $(-3,1)$.

**Explanation:**

Let $A(x_1,y_1)=A(-5,3)$ and $B(x_2,y_2)=B(3,-5)$, $m:n=1:3$

$\therefore x_1=-5, y_1=3, x_2=3, y_2=-5, m=1, n=3$

$\therefore$ By section formula,

$x=\frac{mx_2+nx_1}{m+n}=\frac{1(3)+3(-5)}{1+3}=\frac{3-15}{4}=\frac{-12}{4}=-3$

$y=\frac{my_2+ny_1}{m+n}=\frac{1(-5)+3(3)}{1+3}=\frac{-5+9}{4}=\frac{4}{4}=1$

$\therefore$ Coordinates of P are $(-3,1)$.

**Q10** If the sum of X-coordinates of the vertices of a triangle is $12$ and the sum of Y-coordinates is $9$, then the coordinates of centroid are _____.

(a) $(12,9)$ (b) $(9,12)$ (c) $(4,3)$ (d) $(3,4)$

**ANSWER / SOLUTION**

If the sum of X-coordinates of the vertices of a triangle is $12$ and the sum of Y-coordinates is $9$, then the coordinates of centroid are $(4,3)$.

**Explanation:**

Let $P(x_1,y_1), Q(x_2,y_2), R(x_3,y_3)$ be the coordinates of the vertices of a triangle.

Given that $x_1+x_2+x_3=12$ and $y_1+y_2+y_3=9$

$\therefore$ Coordinates of the centroid $S(x,y)$ are

$x=\frac{x_1+x_2+x_3}{3}=\frac{12}{3}=4$

$y=\frac{y_1+y_2+y_3}{3}=\frac{9}{3}=3$

$\therefore S(x,y)=(4,3)$

## Q.1 (B) One-step Solve

1 Mark each — 10 QUESTIONS · 1 — Revised

**Q1** Find the coordinates of the point of intersection of the graph of the equation $x=2$ and $y=-3$.

**ANSWER / SOLUTION**

The coordinates of the point of intersection of the graph of the equation $x=2$ and $y=-3$ are $(2,-3)$.

**Q2** Find distance between point $A(7,5)$ and $B(2,5)$.

**ANSWER / SOLUTION**

Let $A(x_1,y_1)=A(7,5)$ and $B(x_2,y_2)=B(2,5)$

$\therefore$ By distance formula,

$d(A,B)=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$=\sqrt{(2-7)^2+(5-5)^2}$

$=\sqrt{(-5)^2+0^2}=\sqrt{25}=5$ cm

$\therefore$ The distance between points A and B is $5$ cm.

**Q3** The coordinates of diameter AB of a circle are $A(2,7)$ and $B(4,5)$, then find the coordinates of the centre.

**ANSWER / SOLUTION**

Let $C(x,y)$ be the centre of the circle,

$A(x_1,y_1)=A(2,7), B(x_2,y_2)=B(4,5)$

$\therefore x_1=2, y_1=7, x_2=4, y_2=5$

C is the mid-point of seg AB.

$\therefore$ By midpoint formula,

$C(x,y)=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)=\left(\frac{2+4}{2},\frac{7+5}{2}\right)=\left(\frac{6}{2},\frac{12}{2}\right)=(3,6)$

$\therefore$ The coordinates of the centre of the circle are $(3,6)$.

**Q4** Write the X-coordinate and Y-coordinate of point $P(-5,4)$.

**ANSWER / SOLUTION**

X-coordinate of point $P(-5,4)$ is $-5$.

Y-coordinate of point $P(-5,4)$ is $4$.

**Q5** What are the coordinates of origin?

**ANSWER / SOLUTION**

Coordinates of origin are $(0,0)$.

**Q6** Find distance of point $A(6,8)$ from origin.

**ANSWER / SOLUTION**

Let $A(x_1,y_1)=A(6,8), O(x_2,y_2)=O(0,0)$

$\therefore x_1=6, y_1=8, x_2=0, y_2=0$

By distance formula,

$d(A,O)=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}=\sqrt{(0-6)^2+(0-8)^2}=\sqrt{36+64}=\sqrt{100}=10$

$\therefore$ The distance of point $A(6,8)$ from origin is $10$ units.

**Q7** Find coordinates of midpoint of segment joining $(-2,6)$ and $(8,2)$.

**ANSWER / SOLUTION**

Let $A(x_1,y_1)=A(-2,6), B(x_2,y_2)=B(8,2)$, $C(x,y)$ be the midpoint of seg AB.

$\therefore x_1=-2, y_1=6, x_2=8, y_2=2$

By midpoint formula,

$C(x,y)=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)=\left(\frac{-2+8}{2},\frac{6+2}{2}\right)=\left(\frac{6}{2},\frac{8}{2}\right)=(3,4)$

$\therefore$ Coordinates of midpoint of segment joining $(-2,6)$ and $(8,2)$ are $(3,4)$.

**Q8** Find the coordinates of centroid of a triangle whose vertices are $(4,7)$, $(8,4)$ and $(7,11)$.

TEXTBOOK **Practice Set 5.2 · Q.7 (3) · Page 115**

**ANSWER / SOLUTION**

Let $A(x_1,y_1)=A(4,7), B(x_2,y_2)=B(8,4), C(x_3,y_3)=C(7,11)$

$\therefore$ By centroid formula,

$x=\frac{x_1+x_2+x_3}{3}=\frac{4+8+7}{3}=\frac{19}{3}$

$y=\frac{y_1+y_2+y_3}{3}=\frac{7+4+11}{3}=\frac{22}{3}$

$\therefore$ The coordinates of the centroid are $\left(\frac{19}{3},\frac{22}{3}\right)$.

**Q9** Find distance between points $O(0,0)$ and $B(-5,12)$.

**ANSWER / SOLUTION**

Let $O(x_1,y_1)=O(0,0)$ and $B(x_2,y_2)=B(-5,12)$

$\therefore x_1=0, y_1=0, x_2=-5, y_2=12$

By distance formula,

$d(O,B)=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}=\sqrt{(-5-0)^2+(12-0)^2}=\sqrt{25+144}=\sqrt{169}=13$ units

$\therefore$ The distance between the points O and B is $13$ units.

**Q10** Find coordinates of midpoint of the segment joining points $(0,2)$ and $(12,14)$.

**ANSWER / SOLUTION**

Let $A(x_1,y_1)=A(0,2), B(x_2,y_2)=B(12,14)$

Let the coordinates of the midpoint be $P(x,y)$.

$\therefore$ By midpoint formula,

$x=\frac{x_1+x_2}{2}=\frac{0+12}{2}=6$

$y=\frac{y_1+y_2}{2}=\frac{2+14}{2}=\frac{16}{2}=8$

$\therefore$ The coordinates of the midpoint of the segment joining $(0,2)$ and $(12,14)$ are $(6,8)$.

## Q.2 (A) Activities

2 Marks each

4 QUESTIONS · 2 Revised

**Q1** Find distance between point $Q(3,-7)$ and point $R(3,3)$.

**Solution:** Suppose $Q(x_1,y_1)$ and point $R(x_2,y_2)$

$x_1=3, y_1=-7$ and $x_2=3, y_2=3$

**Using distance formula,**

$d(Q,R)=\sqrt{\square}$

$\therefore d(Q,R)=\sqrt{\square-100}$

$\therefore d(Q,R)=\sqrt{\square}$

$\therefore d(Q,R)=\square$

**ANSWER / SOLUTION**

Suppose $Q(x_1,y_1)$ and point $R(x_2,y_2)$

$x_1=3, y_1=-7$ and $x_2=3, y_2=3$

Using distance formula,

$d(Q,R)=\sqrt{\boxed{(x_2-x_1)^2+(y_2-y_1)^2}}$

$=\sqrt{(3-3)^2+[3-(-7)]^2}$

$=\sqrt{0^2+(10)^2}$

$\therefore d(Q,R)=\sqrt{\boxed{0}+100}$

$\therefore d(Q,R)=\sqrt{\boxed{100}}$

$\therefore d(Q,R)=\boxed{10}$

**Q2** **Find distance between point $A(-1,1)$ and point $B(5,-7)$:**

**Solution: Suppose $A(x_1,y_1)$ and $B(x_2,y_2)$**

**$x_1=-1, y_1=1$ and $x_2=5, y_2=-7$**

**Using distance formula,**

$d(A,B)=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$\therefore d(A,B)=\sqrt{\square+[(-7)+\square]^2}$

$\therefore d(A,B)=\sqrt{\square}$

$\therefore d(A,B)=\square$

**TEXTBOOK Solved Example · Ex. (1) · Page 104**

**ANSWER / SOLUTION**

Suppose $A(x_1,y_1)$ and $B(x_2,y_2)$

$x_1=-1, y_1=1$ and $x_2=5, y_2=-7$

Using distance formula,

$d(A,B)=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$\therefore d(A,B)=\sqrt{\boxed{[5-(-1)]^2}+[(-7)+\boxed{(-1)}]^2}$

$\therefore d(A,B)=\sqrt{6^2+(-8)^2}=\sqrt{36+64}$

$\therefore d(A,B)=\sqrt{\boxed{100}}$

$\therefore d(A,B)=\boxed{10 \text{ units}}$

**Q3** **Find coordinates of the midpoint of a segment joining point $A(-1,1)$ and point $B(5,-7)$**

**Solution: Suppose $A(x_1,y_1)$ and $B(x_2,y_2)$**

**$x_1=-1, y_1=1$ and $x_2=5, y_2=-7$**

**Using midpoint formula,**

**$\therefore$ Coordinates of midpoint of segment AB**

$=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$

$=\left(\frac{\square}{2},\frac{\square}{2}\right)$

**$\therefore$ Coordinates of the midpoint $=\left(\frac{4}{2},\frac{\square}{2}\right)$**

**$\therefore$ Coordinates of the midpoint $=(2,\square)$**

**ANSWER / SOLUTION**

Suppose $A(x_1,y_1)$ and $B(x_2,y_2)$

$x_1=-1, y_1=1$ and $x_2=5, y_2=-7$

Using midpoint formula,

$\therefore$ Coordinates of midpoint of segment AB

$=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$

$=\left(\frac{\boxed{-1+5}}{2},\frac{\boxed{1-7}}{2}\right)$

$\therefore$ Coordinates of the midpoint $=\left(\frac{4}{2},\frac{\boxed{-6}}{2}\right)$

$\therefore$ Coordinates of the midpoint $=(2,\boxed{-3})$

**Q4** The coordinates of the vertices of a triangle ABC are $A(-7, 6)$, $B(2, -2)$ and $C(8, 5)$. Find coordinates of its centroid.

**Solution:** Suppose $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$

$x_1 = -7, y_1 = 6$ and $x_2 = 2, y_2 = -2$ and $x_3 = 8, y_3 = 5$

Using Centroid formula,

∴ Coordinates of the centroid of a triangle ABC

$= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

$= \left(\frac{\square}{3}, \frac{\square}{3}\right)$

∴ Coordinates of the centroid of a triangle ABC $= \left(\frac{3}{3}, \square\right)$

∴ Coordinates of the centroid of a triangle ABC $= (1, \square)$

TEXTBOOK **Practice Set 5.2 · Q.7 (1) · Page 115**

**ANSWER / SOLUTION**

Suppose $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$

$x_1 = -7, y_1 = 6$ and $x_2 = 2, y_2 = -2$ and $x_3 = 8, y_3 = 5$

Using Centroid formula,

∴ Coordinates of the centroid of a triangle ABC

$= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

$= \left(\frac{\boxed{-7+2+8}}{3}, \frac{\boxed{6-2+5}}{3}\right)$

∴ Coordinates of the centroid of a triangle ABC $= \left(\frac{3}{3}, \frac{\boxed{9}}{3}\right)$

∴ Coordinates of the centroid of a triangle ABC $= (1, \boxed{3})$

## Q.2 (B) Short Solve

2 Marks each

5 QUESTIONS · 4 ☐ Revised

**Q1** The point Q divides segment joining $A(3, 5)$ and $B(7, 9)$ in the ratio $2 : 3$. Find the X-coordinate of Q.

TEXTBOOK **Solved Example · Ex. (1) (x-co-ordinate) · Page 110**

**ANSWER / SOLUTION**

Let the coordinates of point Q be $(x, y)$ and $A(x_1, y_1), B(x_2, y_2)$ be the given points.

Here, $x_1 = 3, y_1 = 5, x_2 = 7, y_2 = 9, m = 2, n = 3$

∴ By section formula,

$x = \frac{mx_2+nx_1}{m+n} = \frac{2(7)+3(3)}{2+3} = \frac{14+9}{5} = \frac{23}{5}$

$y = \frac{my_2+ny_1}{m+n} = \frac{2(9)+3(5)}{2+3} = \frac{18+15}{5} = \frac{33}{5}$

∴ The coordinates of point Q are $\left(\frac{23}{5}, \frac{33}{5}\right)$.

∴ The X-coordinate of Q is $\frac{23}{5}$.

**Q2** If the distance between point $L(x, 7)$ and point $M(1, 15)$ is $10$, then find the value of $x$.

TEXTBOOK **Practice Set 5.1 · Q.7 · Page 108**

**ANSWER / SOLUTION**

Let $L(x_1, y_1) = L(x, 7)$ and $M(x_2, y_2) = M(1, 15)$

$x_1 = x, y_1 = 7, x_2 = 1, y_2 = 15$

By distance formula,

$d(L, M) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$\therefore 10 = \sqrt{(1 - x)^2 + (15 - 7)^2}$

$\therefore 10 = \sqrt{(1 - x)^2 + 8^2}$

$\therefore 100 = (1 - x)^2 + 64$ ---[Squaring both sides]

$\therefore (1 - x)^2 = 36$

$\therefore 1 - x = \pm\sqrt{36} \implies 1 - x = \pm 6$

$\therefore 1 - x = 6$ or $1 - x = -6$

$\therefore x = -5$ or $x = 7$

$\therefore$ The value of $x$ is $-5$ or $7$.

**Q3** Find the coordinates of midpoint of segment joining $(22, 20)$ and $(0, 16)$.

TEXTBOOK **Practice Set 5.2 · Q.6 · Page 115**

**ANSWER / SOLUTION**

Let $A(x_1, y_1) = A(22, 20), B(x_2, y_2) = B(0, 16)$

Let the coordinates of the midpoint be $P(x, y)$.

$\therefore$ By midpoint formula,

$x = \frac{x_1 + x_2}{2} = \frac{22 + 0}{2} = 11$

$y = \frac{y_1 + y_2}{2} = \frac{20 + 16}{2} = \frac{36}{2} = 18$

$\therefore$ The coordinates of the midpoint of the segment joining $(22, 20)$ and $(0, 16)$ are $(11, 18)$.

**Q4** Find distance CD where $C(-3a, a), D(a, -2a)$.

TEXTBOOK **Problem Set 5 · Q.6 (iii) · Page 122**

**ANSWER / SOLUTION**

Let $C(x_1, y_1)$ and $D(x_2, y_2)$ be the given points.

$\therefore x_1 = -3a, y_1 = a, x_2 = a, y_2 = -2a$

By distance formula,

$d(C, D) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{[a - (-3a)]^2 + (-2a - a)^2}$

$= \sqrt{(a + 3a)^2 + (-3a)^2}$

$= \sqrt{(4a)^2 + (-3a)^2}$

$= \sqrt{16a^2 + 9a^2} = \sqrt{25a^2} = 5a$ units

$\therefore d(C, D) = 5a$ units

**Q5** Show that the point $(11, -2)$ is equidistant from $(4, -3)$ and $(6, 3)$.

**ANSWER / SOLUTION**

Let $P(x_1, y_1) = P(11, -2), Q(x_2, y_2) = Q(4, -3), R(x_3, y_3) = R(6, 3)$

By distance formula,

$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(4 - 11)^2 + [-3 - (-2)]^2} = \sqrt{(-7)^2 + (-1)^2} = \sqrt{49 + 1} = \sqrt{50} = 5\sqrt{2}$

And,

$d(P, R) = \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} = \sqrt{(6 - 11)^2 + [3 - (-2)]^2} = \sqrt{(-5)^2 + (5)^2} = \sqrt{25 + 25} = \sqrt{50} = 5\sqrt{2}$

Here, $d(P, Q) = d(P, R)$

$\therefore$ Point $(11, -2)$ is equidistant from $(4, -3)$ and $(6, 3)$.

## Q.3 (A) Activities

3 Marks each

2 QUESTIONS · 1 · Revised

**Q1** If the point $P(6, 7)$ divides the segment joining $A(8, 9)$ and $B(1, 2)$ in some ratio, find that ratio.

**Solution:** Point P divides segment AB in the ratio $m : n$.

$A(8, 9) = (x_1, y_1), B(1, 2) = (x_2, y_2)$ and $P(6, 7) = (x, y)$

Using Section formula of internal division,

$\therefore 7 = \frac{m(\square) - n(9)}{m + n}$

$\therefore 7m + 7n = \square + 9n$

$\therefore 7m - \square = 9n - \square$

$\therefore \square = 2n$

$\therefore \frac{m}{n} = \square$

**ANSWER / SOLUTION**

Point P divides segment AB in the ratio $m : n$.

$A(8, 9) = (x_1, y_1), B(1, 2) = (x_2, y_2)$ and $P(6, 7) = (x, y)$

Using Section formula of internal division,

$y = \frac{my_2 + ny_1}{m + n}$

$\therefore 7 = \frac{m(\boxed{2}) + n(9)}{m + n}$

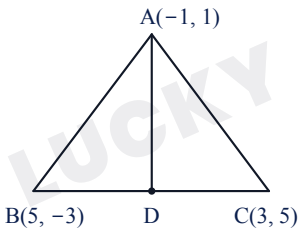
$\therefore 7m + 7n = \boxed{2m} + 9n$

$\therefore 7m - \boxed{2m} = 9n - \boxed{7n}$

$\therefore \boxed{5m} = 2n$

$\therefore \frac{m}{n} = \boxed{\frac{2}{5}}$

**Q2** From the figure given alongside, find the length of the median AD of triangle ABC. Complete the activity.




**Solution:** Here $A(-1,1), B(5,-3), C(3,5)$ and suppose $D(x,y)$ are coordinates of point D.

Using midpoint formula,

$x = \frac{5+3}{2} \implies x = \square$

$y = \frac{-3+5}{2} \implies y = \square$

Using distance formula,

$\therefore AD = \sqrt{(4-\square)^2 + (1-1)^2}$

$\therefore AD = \sqrt{(\square)^2 + (0)^2}$

$\therefore AD = \sqrt{\square}$

$\therefore$ The length of median AD $= \square$

TEXTBOOK **Problem Set 5 · Q.13 (median AD) · Page 123**

**ANSWER / SOLUTION**

Here $A(-1,1), B(5,-3), C(3,5)$ and suppose $D(x,y)$ are coordinates of point D. D is the midpoint of seg BC.

Using midpoint formula,

$x = \frac{x_1+x_2}{2} = \frac{5+3}{2} = \frac{8}{2} \implies x = \boxed{4}$

$y = \frac{y_1+y_2}{2} = \frac{-3+5}{2} = \frac{2}{2} \implies y = \boxed{1}$

Using distance formula,

$\therefore AD = \sqrt{(4-\boxed{-1})^2 + (1-1)^2}$

$\therefore AD = \sqrt{(\boxed{5})^2 + (0)^2}$

$\therefore AD = \sqrt{\boxed{25}}$

$\therefore$ The length of median AD $= \boxed{5 \text{ units}}$

## Q.3 (B) Solve / Prove

3 Marks each

7 QUESTIONS · 3 ☐ Revised

**Q1** Show that $P(-2,2)$, $Q(2,2)$ and $R(2,7)$ are vertices of a right angled triangle.

TEXTBOOK **Practice Set 5.1 · Q.4 · Page 107**

**ANSWER / SOLUTION**

Distance between two points $= \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

By distance formula,

$PQ = \sqrt{[2-(-2)]^2 + (2-2)^2} = \sqrt{(2+2)^2 + 0^2} = \sqrt{16} = 4$ ---(i)

$QR = \sqrt{(2-2)^2 + (7-2)^2} = \sqrt{0^2 + 5^2} = \sqrt{25} = 5$ ---(ii)

$PR = \sqrt{[2-(-2)]^2 + (7-2)^2} = \sqrt{4^2 + 5^2} = \sqrt{16+25} = \sqrt{41}$

Now, $PR^2 = (\sqrt{41})^2 = 41$ ---(iii)

Consider, $PQ^2 + QR^2 = 4^2 + 5^2 = 16 + 25 = 41$ ---[From (i) and (ii)]

$\therefore PR^2 = PQ^2 + QR^2$ ---[From (iii)]

$\therefore \Delta PQR$ is a right angled triangle. ---[Converse of Pythagoras theorem]

$\therefore$ Points P, Q, and R are the vertices of a right angled triangle.

**Q2** Show that the point $(0, 9)$ is equidistant from the points $(-4, 1)$ and $(4, 1)$.

**ANSWER / SOLUTION**

Let $P(x_1, y_1) = P(0, 9), Q(x_2, y_2) = Q(-4, 1), R(x_3, y_3) = R(4, 1)$

By distance formula,

$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{[(-4) - 0]^2 + (1 - 9)^2} = \sqrt{(-4)^2 + (-8)^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}$

And,

$d(P, R) = \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} = \sqrt{(4 - 0)^2 + (1 - 9)^2} = \sqrt{4^2 + (-8)^2} = \sqrt{16 + 64} = \sqrt{80} = 4\sqrt{5}$

Here, $d(P, Q) = d(P, R)$

$\therefore$ The point $(0, 9)$ is equidistant from $(-4, 1)$ and $(4, 1)$.

**Q3** Point $P(-4, 6)$ divides point $A(-6, 10)$ and $B(m, n)$ in the ratio $2 : 1$, then find the coordinates of point B.

TEXTBOOK **Solved Example · Ex. (2) · Page 113**

**ANSWER / SOLUTION**

By section formula,

$-4 = \frac{2 \times m + 1 \times (-6)}{2+1}$

$\therefore -4 = \frac{2m-6}{3} \implies -12 = 2m - 6 \implies 2m = -6 \implies m = -3$

Also,

$6 = \frac{2 \times n + 1 \times 10}{2+1}$

$\therefore 6 = \frac{2n+10}{3} \implies 18 = 2n + 10 \implies 2n = 8 \implies n = 4$

$\therefore$ Coordinates of point B are $(-3, 4)$.

**Q4** Show that points $A(-4, -7)$, $B(-1, 2)$, $C(8, 5)$ and $D(5, -4)$ are the vertices of a parallelogram ABCD.

TEXTBOOK **Practice Set 5.3 · Q.5 · Page 121**

**ANSWER / SOLUTION**

We know that, $\text{Slope of line} = \frac{y_2 - y_1}{x_2 - x_1}$

$\text{Slope of side AB} = \frac{2-(-7)}{-1-(-4)} = \frac{2+7}{-1+4} = \frac{9}{3} = 3$ ---(i)

$\text{Slope of side BC} = \frac{5-2}{8-(-1)} = \frac{3}{8+1} = \frac{3}{9} = \frac{1}{3}$ ---(ii)

$\text{Slope of side CD} = \frac{-4-5}{5-8} = \frac{-9}{-3} = 3$ ---(iii)

$\text{Slope of side AD} = \frac{-4-(-7)}{5-(-4)} = \frac{-4+7}{5+4} = \frac{3}{9} = \frac{1}{3}$ ---(iv)

$\therefore$ Slope of side AB = Slope of side CD ---[From (i) and (iii)]

$\therefore$ side AB $\parallel$ side CD

$\therefore$ Slope of side BC = Slope of side AD ---[From (ii) and (iv)]

$\therefore$ side BC $\parallel$ side AD

Both the pairs of opposite sides of ABCD are parallel.

$\therefore$ Points $A(-4, -7)$, $B(-1, 2)$, $C(8, 5)$ and $D(5, -4)$ are the vertices of a parallelogram.

**Q5** **Show that the points $(0,-1)$, $(8,3)$, $(6,7)$ and $(-2,3)$ are vertices of a rectangle.**

**ANSWER / SOLUTION**

Let the points be $P(0,-1), Q(8,3), R(6,7), S(-2,3)$

Distance between two points $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$\therefore d(P,Q) = \sqrt{(8-0)^2+[3-(-1)]^2} = \sqrt{8^2+4^2} = \sqrt{64+16} = \sqrt{80}$ ---(i)

$d(Q,R) = \sqrt{(6-8)^2+(7-3)^2} = \sqrt{(-2)^2+4^2} = \sqrt{4+16} = \sqrt{20}$ ---(ii)

$d(R,S) = \sqrt{[(-2)-6]^2+(3-7)^2} = \sqrt{(-8)^2+(-4)^2} = \sqrt{64+16} = \sqrt{80}$ ---(iii)

$d(P,S) = \sqrt{[(-2)-0]^2+[3-(-1)]^2} = \sqrt{(-2)^2+4^2} = \sqrt{4+16} = \sqrt{20}$ ---(iv)

In $\square PQRS$,

side $PQ =$ side $RS$ ---[From (i) and (iii)]

side $QR =$ side $PS$ ---[From (ii) and (iv)]

$\therefore \square PQRS$ is a parallelogram.

Now, diagonals:

$d(P,R) = \sqrt{(6-0)^2+[7-(-1)]^2} = \sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100} = 10$ ---(v)

$d(Q,S) = \sqrt{[(-2)-8]^2+(3-3)^2} = \sqrt{(-10)^2+0^2} = \sqrt{100} = 10$ ---(vi)

In parallelogram PQRS, diagonal $PR =$ diagonal $QS$ ---[From (v) and (vi)]

$\therefore \square PQRS$ is a rectangle.

**Q6** **Show that the points $(2,0)$, $(-2,0)$ and $(0,2)$ are vertices of a triangle. State the type of triangle with reason.**

**ANSWER / SOLUTION**

Let the points be $P(2,0), Q(-2,0)$ and $R(0,2)$.

By distance formula,

$d(P,Q) = \sqrt{[(-2)-2]^2+(0-0)^2} = \sqrt{(-4)^2+0} = \sqrt{16} = 4$ ---(i)

$d(Q,R) = \sqrt{[0-(-2)]^2+(2-0)^2} = \sqrt{2^2+2^2} = \sqrt{4+4} = \sqrt{8}$ ---(ii)

$d(P,R) = \sqrt{(0-2)^2+(2-0)^2} = \sqrt{(-2)^2+2^2} = \sqrt{4+4} = \sqrt{8}$ ---(iii)

On adding (ii) and (iii), $d(Q,R)+d(P,R) = \sqrt{8}+\sqrt{8} = 2\sqrt{8} = 2(2\sqrt{2}) = 4\sqrt{2} > 4 = d(P,Q)$

$\therefore$ Points P, Q, R are non-collinear points.

We can construct a triangle through 3 non-collinear points.

Since $d(Q,R) = d(P,R) = \sqrt{8}$,

$\therefore \Delta PQR$ is an isosceles triangle.

$\therefore$ The segment joining the points $(2,0)$, $(-2,0)$ and $(0,2)$ will form an isosceles triangle.

**Q7** **If $A(5,4)$, $B(-3,-2)$ and $C(1,-8)$ are the vertices of a $\Delta ABC$. Segment AD is median. Find the length of seg AD.**

**ANSWER / SOLUTION**

Since segment AD is median, Point D is the midpoint of side BC.

By midpoint formula,

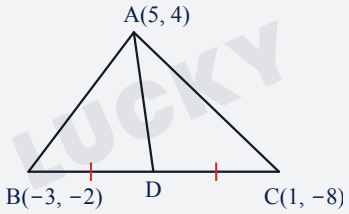
Coordinates of D $= \left(\frac{-3+1}{2}, \frac{-2-8}{2}\right) = \left(\frac{-2}{2}, \frac{-10}{2}\right) = (-1,-5)$

Distance between two points $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

By distance formula,

$d(A,D) = \sqrt{[5-(-1)]^2+[4-(-5)]^2} = \sqrt{(5+1)^2+(4+5)^2} = \sqrt{6^2+9^2} = \sqrt{36+81} = \sqrt{117}$

$\therefore$ The length of segment AD is $\sqrt{117}$ units.



## Q.4 Long Solve / Prove 4 Marks each

3 QUESTIONS · 1 ☐ Revised

**Q1** Show that $A(1,2)$, $B(1,6)$, $C(1+2\sqrt{3},4)$ are vertices of an equilateral triangle.

TEXTBOOK Practice Set 5.1 · Q.8 · Page 108

**ANSWER / SOLUTION**

Distance between two points $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

By distance formula,

$AB = \sqrt{(1-1)^2+(6-2)^2} = \sqrt{0^2+4^2} = \sqrt{16} = 4$ ---(i)

$BC = \sqrt{(1+2\sqrt{3}-1)^2+(4-6)^2} = \sqrt{(2\sqrt{3})^2+(-2)^2} = \sqrt{12+4} = \sqrt{16} = 4$ ---(ii)

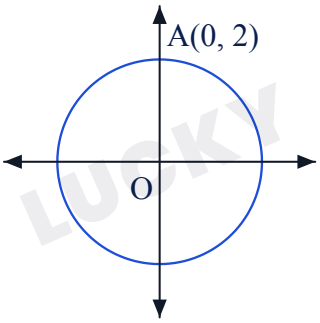
$AC = \sqrt{(1+2\sqrt{3}-1)^2+(4-2)^2} = \sqrt{(2\sqrt{3})^2+2^2} = \sqrt{12+4} = \sqrt{16} = 4$ ---(iii)

$\therefore AB = BC = AC = 4$ ---[From (i), (ii) and (iii)]

$\therefore \Delta ABC$ is an equilateral triangle.

$\therefore$ Points A, B and C are the vertices of an equilateral triangle.

**Q2** Seg OA is the radius of a circle with centre O. The coordinates of point A is $(0,2)$ then decide whether the point $B(1,2)$ is on the circle?




**ANSWER / SOLUTION**

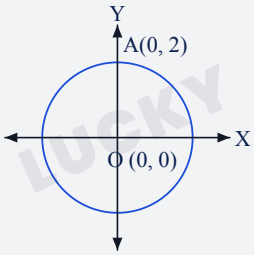
Coordinates of A $= (0,2)$

Coordinates of O $= (0,0)$

Coordinates of B $= (1,2)$

By distance formula,

$d(O,A) = \sqrt{(0-0)^2+(0-2)^2} = \sqrt{0+(-2)^2} = \sqrt{4} = 2$ ---(i)




$d(O,B) = \sqrt{(0-1)^2+(0-2)^2} = \sqrt{(-1)^2+(-2)^2} = \sqrt{1+4} = \sqrt{5}$ ---(ii)

$\therefore$ From (i) and (ii),

$d(O,B) > d(O,A)$

$\therefore d(O,B) >$ Radius of circle

$\therefore$ Point $B(1,2)$ does not lie on the circle but lies outside the circle.

**Q3** Find the ratio in which Y-axis divides the point $A(3,5)$ and point $B(-6,7)$. Find the coordinates of the point.

**ANSWER / SOLUTION**

Let C be a point on Y-axis which divides seg AB in the ratio $m:n$.

Point C lies on the Y-axis.

$\therefore$ Its X-coordinate is $0$.

Let $C = (0,y)$

Here, $A(x_1,y_1) = A(3,5), B(x_2,y_2) = B(-6,7)$

By Section formula,

$x = \frac{mx_2+nx_1}{m+n}$

$\therefore 0 = \frac{-6m+3n}{m+n} \implies -6m+3n = 0 \implies 6m = 3n \implies \frac{m}{n} = \frac{3}{6} = \frac{1}{2}$

$\therefore m:n = 1:2$ ---(i)

Now for Y-coordinate:

$y = \frac{my_2+ny_1}{m+n} = \frac{1(7)+2(5)}{1+2} = \frac{7+10}{3} = \frac{17}{3}$

$\therefore$ Y-axis divides the seg AB in the ratio $1:2$ and the coordinates of that point are $\left(0, \frac{17}{3}\right)$.

## Q.5 Creative / HOTS 3 Marks each

2 QUESTIONS ☐ Revised

**Q1** The points $(7,-6)$, $(2,k)$ and $(h,18)$ are the vertices of triangle. If $(1,5)$ are the coordinates of centroid, find the value of h and k.

**ANSWER / SOLUTION**

Let $(7,-6) \equiv (x_1,y_1)$, $(2,k) \equiv (x_2,y_2)$, $(h,18) \equiv (x_3,y_3)$ be the three vertices of the triangle.

$(1,5)$ are coordinates of centroid.

$\therefore$ By Centroid formula,

$1 = \frac{x_1+x_2+x_3}{3} \implies 1 = \frac{7+2+h}{3} \implies 3 = 9+h \implies h = -6$

$5 = \frac{y_1+y_2+y_3}{3} \implies 5 = \frac{-6+k+18}{3} \implies 15 = 12+k \implies k = 3$

$\therefore h = -6, k = 3$

**Q2** Using distance formula decide whether the points $(4,3)$, $(5,1)$ and $(1,9)$ are collinear or not.

**ANSWER / SOLUTION**

Let $A(x_1,y_1) = A(4,3), B(x_2,y_2) = B(5,1), C(x_3,y_3) = C(1,9)$

$\therefore d(A,B) = \sqrt{(5-4)^2+(1-3)^2} = \sqrt{1^2+(-2)^2} = \sqrt{1+4} = \sqrt{5}$ ---(i)

$d(B,C) = \sqrt{(1-5)^2+(9-1)^2} = \sqrt{(-4)^2+8^2} = \sqrt{16+64} = \sqrt{80} = 4\sqrt{5}$ ---(ii)

$d(A,C) = \sqrt{(1-4)^2+(9-3)^2} = \sqrt{(-3)^2+6^2} = \sqrt{9+36} = \sqrt{45} = 3\sqrt{5}$ ---(iii)

Now, $d(A,B)+d(A,C) = \sqrt{5}+3\sqrt{5} = 4\sqrt{5} = d(B,C)$ ---[From (i), (ii) and (iii)]

$\therefore$ Points A, B, C are collinear.

**Practice daily. Plot every point. Score full marks.** ⚡
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